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Modelling economic policy issues

Does innovation affect the impact of corruption on economic growth? International evidence

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ABSTRACT

The corruption-economic growth nexus and the relationship between innovation and corruption have long been extensively studied. While individual issues have been empirically analysed, two effects have not yet been sufficiently researched: (1) the overall effect of innovation on the corruption-growth relationship and (2) the direction of causality between these variables. The present article used data from 109 countries over the 2010–2018 period and focused on the macro level. We employed Granger causality tests for panel data (Dumitrescu–Hurlin) to determine the direction of causation among variables. We also employed nonstationary panel techniques with Fully Modified Ordinary Least Square (FM-OLS) to assess stationarity and long-run relationships. After investigating the direct impact of corruption on economic growth, we then examined the indirect effect through different transmission channels, including innovation. The results were tested and found robust in subcategories based on the developmental and regional levels. Finally, this study analysed the impact of innovation on the relationship between corruption and economic growth. The results indicated a robust negative relationship between corruption and economic growth and corruption and innovation. The findings also revealed a positive relationship between innovation and economic growth and a bi-directional causality between the variables. Furthermore, innovation was found to reduce the harmful effects of corruption on economic growth, mainly in developed countries.

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1. Introduction

In the empirical research, there is a lack of consensus on how corruption affects economic growth, albeit primarily in theory. According to the "grease the wheels" hypothesis, corruption stimulates economic growth by overcoming inefficient bureaucracies. Businesspeople, politicians, and administrators are likely to drive growth through corruption when regulations affecting economic activity (e.g., investment approval, business start-ups, licence for patents and trademarks) are stringent (Leff, 1964; Acemoglu and Verdier, 2000; Huang, 2015). In contrast, the "sand in the wheels" hypothesis posits that corruption reduces economic growth by preventing efficient production and innovation (Mauro, 1995; Mo, 2001; Aidt et al., 2008); (Cieślik and Goczek, 2018).

While the theory of economic growth and empirical research recognise that corruption can drastically affect the stable conditions under which a country's economy develops (Hannafey, 2003), the effects of corruption on innovation and

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entrepreneurship are less understood. It is also noteworthy that the impact of innovation on corruption and economic growth is complex, bidirectional, and varies from one nation to another. Moreover, empirical research is limited, and the results are unclear. This article aims to shed light on these issues.

Corruption is known to have a corrosive impact on any country's economic growth, which is why it is important to undertake a thorough investigation of the linkages between the two via an empirical model. The next step entails exploring how corruption impacts innovative activities. According to the extant literature, corruption negatively affects innovative activities because innovators need government-provided goods more than ordinary producers (Mo, 2001; Xu and Yano, 2017; Dincer, 2019). Therefore, they become primary targets of corruption. In addition, innovators usually do not have their own resources (as they need extensive funding) for their investments, making them vulnerable to corruption (Murphy et al., 1993; Anokhin and Schulze, 2009). As a result, their skills and efforts are directed to corrupt processes rather than productive investments. Finally, to complete the study of the dynamic correlation between the three variables, we must investigate how innovation intervenes in the corruption-growth nexus, through what processes, and the extent of its impact. However, for that to transpire, an appropriate sample and econometric methodologies are required. In addition to employing Panel-Corrected Standard Errors (PCSE) models and cointegration analysis through the econometric dynamic FM-OLS method, we used causation analysis with the Granger causality processes. This was necessary to investigate the degree of correlation, the effects of the variables, and the direction of their causality.

As per empirical research, a one-unit increase in corruption reduces GDPpc by 1.05% to 1.55%. However, the indirect effects of corruption through the transmission channels account for just 31.16% of the total impact. This overall result is primarily attributed to the impact of corruption on economic growth through human capital, total investment, and innovation, corresponding to 39.52%, 9.44%, and 19.88%, respectively.

Improved innovation is known to cause a significant improvement in per capita GDP and reduce the harmful effects of corruption on economic growth. Beyond a particular threshold, the impact of corruption is reversed. This threshold varies when trademarks and patents are used but remains significant. These findings reiterate the need to fight corruption. Policymakers should pursue administrative reforms to promote transparency, efficiency, and fair competition. Our findings show that great emphasis needs to be placed on strengthening innovation, which, in turn, reduces the effects of corruption by lowering demand and, ultimately, supply.

Our research contributes to the literature in various ways. First, previous studies have used individual companies, states, or a small sample of countries (Mauro, 1995; Goedhuys et al., 2016; Ogun, 2018; Dincer, 2019; Das et al., 2020b; Ellis et al., 2020). Conversely, this study utilises a large sample with reliable measurements: 109 countries from 2010–2018. This period was chosen because it corresponds to post-crisis years, and reliable data was presented on corruption and innovation indicators. We also utilised the control of corruption index (CCI), the international country risk guide (ICRG), and the corruption perception index (CPI), along with innovation measurements with reliable Research and Development (R&D) expenses (%GDP), trademarks, and patents. The research focuses on the effect of variables at the macro level, investigating the general interactions of the variables at the country level. It also uses corruption measurements related to general trends (perceptions). Robustness checks were also carried out, dividing our overall model into developmental categories and regions, drawing geographical consistency conclusions and reducing the estimation bias.

Second, we endeavour to clarify the interactions between corruption, innovation, and economic growth. We first examine how corruption affects innovation and how that impact is transmitted to economic growth. Direct and indirect procedures (transmission channels) clarify how corruption influences growth. Prior studies have examined the indirect effects mainly through the channels of human capital, investment, and trade freedom (Mo, 2001; Pellegrini and Gerlagh, 2004; Gründler and Potrafke, 2019). Our research contributes to the innovation channel, which we control with various aspects through the variables used. Through these transmission channels, one of the main objectives of the research is to assess the degree of the impact of corruption on innovation transferred to GDP per capita, a process that has yet to be analysed. Third, we explore in which countries and under what conditions innovation dilutes the harmful effects of corruption on economic growth. Previous studies have investigated the simultaneous effects of corruption and innovation on a model of one country (Ogun, 2018), a sample of countries (Das et al., 2020b), or on the firm level by estimating micro-level interactions (Goedhuys et al., 2016). By contrast, this study introduces the correlated effect of innovation on the corruption-growth nexus through cross-term variables. More specifically, we explore how this effect changes, the thresholds of influence, and the countries where this effect is most pronounced.

Finally, we use the PCSE econometric methodology, which deals with the problem of cross-sectional dependence and allows us to use the causality procedures (Hoechle, 2007). In addition, we use the FM-OLS methodology following Pedroni (2001), which effectively addresses the problem of both endogenous and omitted variables (Roodman, 2009; Dincer, 2019) and cross-sectional dependence (Baltagi, 2006; Costantini and Destefanis, 2009). As part of these processes, many different measurements of corruption and innovation were used, and different econometric models were incorporated. All processes are confirmed by robustness analysis.

The structure of this paper is organised as follows: Section 2 covers economic growth, corruption, governance, innovation, and the relationship between them according to the literature. Section 3 discusses the research methodology and data. Section 4 presents the econometric tests, the empirical results, and the consequent discussion. Finally, Section 5 provides concluding remarks and suggestions for future research.

2. Literature review

2.1. The relationship between corruption and economic growth

Theoretical studies

Two theories describe the relationship between corruption and economic growth: "grease the wheels" and "sand the wheels". Leff (1964) and Summers (1977) posited that corruption has a positive effect on economic growth, such that (1) it bypasses bureaucracy and (2) it encourages corrupt government officials to work more and more efficiently. On the other hand, Myrdal (1968) identified corrupt civil servants as causing further delays. Meanwhile, Rose-Ackerman (1978) opined that it is difficult to reduce corruption in areas where economic conditions favour it. According to Shleifer and Vishny (1993), corruption deters economic growth.

Several researchers have theoretically confirmed the negative impact of corruption on the state economy and described events, thereby confirming the harmfulness of such an influence (Blackburn et al., 2006; De Vaal and Ebben, 2011; Ivanyna et al., 2016). According to Blackburn et al. (2006), the negative impact of corruption on the country's economy and, in particular, on its productivity has been identified, where the authors argue that different countries have different productivity levels. Thus, it explicates the differences in corruption's effects on these countries' economies (Haque and Kneller, 2009; Akimova et al., 2020). Also, these latter studies determine the limit of corruption levels. Before reaching this limit, researchers contend that the hypothesis of "grease the wheels", according to which corruption can positively impact economic growth, is possible. Through this process, governments have the opportunity to know if they need to take action to combat corruption and under what conditions.

Some researchers followed a different theoretical methodology. They tried to show the indirect effect of corruption on economic growth by examining corruption as a cause of uncertainty. Obstfeld (1994) and Devereux and Smith (1994) studied the effects of risk dispersion on growth models in open economies. These models investigate the investment trend in different economies depending on the uncertainty of investments or their variety concerning the uncertainty that exists when unbalanced factors are introduced (e.g., corruption). The uncertainty on economic growth caused by corruption was calculated mathematically. Abadie and Gardeazabal's (2008) approach is more prevalent, as they have reduced the variables to stochastic equations and introduced corruption as a Poisson stochastic process. This model investigates how corruption affects the expected return on investment and is associated with increased uncertainty, which facilitates the evaluation of countries and the corresponding preferences of investors, depending on the corruption index (Raddatz and Schmukler, 2012). By adopting a similar procedure, Cieślik and Goczek (2018) found that the corruption-induced ambiguity in the equations acts as a tax on the transaction process. This reduces the return on investment and increases its volatility, which inexorably instils doubts in investors. Thus, it becomes evident that corruption directly hinders economic growth by deterring investment.

In this context, Bologna Pavlik (2018) investigates the causes of the fluctuations in the results of corruption on economic growth, examining the relationship between corruption and economic growth that incorporates the uncertainty around bribery payments in the regression. More specifically, bribery is found to positively affect growth (grease the wheels) at very low levels of uncertainty. However, after overcoming a certain level of uncertainty, bribery negatively affects growth (sand in the wheels).

Finally, Akimova et al. (2020) provided theoretical evidence for the countries' corruption events that lead to economic strain. They investigated that corruption in different countries has varied intents of corruption, but the comprehensiveness of such corrupt acts is proven. They evaluated the impact of corruption on the economic security of different states. Notably, Akimova et al. (2020) analysed corruption levels in the form of graphs using a comparison method, which is a graphical analysis and synthesis way of knowing about the target goal to determine the negative impact of corruption.

Empirical studies

The voluminous empirical research on the correlation between corruption and economic growth generally follows three directions. First, some studies have identified the positive effects of corruption on economic growth. The "grease the wheels" hypothesis finds a positive effect of corruption on economic growth under certain conditions (Acemoglu and Verdier, 2000). The same conclusion was reached by Huang (2015), who used data from Asian and Pacific countries for the period 1997–2013, and showed a positive correlation between the two variables.

The second strand of literature focuses on the negative relationship between corruption and economic growth. Examining the relationship between growth and investment, Barro (1991) found that corruption reduces growth indirectly through investment. Mauro (1995) used data from 67 countries to identify a negative correlation between corruption and the average annual economic growth rate. Davoodi and Tanzi (1997) also investigated the effects of corruption on the economic size of countries and found that it prioritises public investment over private investment by strongly substituting productive capital for the economy. Fayissa and Nsiah (2013) concluded that income level plays an essential role in studying the impact of governance on economic growth. Spector (2016) affirmed that corruption could be combated with a combination of solid institutions, a sound legal body, and a clear political will. Reliable economic institutions and reducing corruption also improve growth prospects. Malanski and Póvoa (2021) acknowledged that institutional quality affects corrupt activities, given that the presence of transparent and credible institutions discourages such activities, improving the pace of economic growth.

Research has also been conducted linking corruption and economic growth to other factors such as public debt, taxes, and e-government effectiveness. Krishnan et al. (2013) explored the effects of corruption and e-government maturity on economic prosperity and growth. Hypotheses were tested via a cross-sectional analysis of 105 countries from 2004 to 2008. They provided a multivariable model using urbanity, manufacturing, population, exports, environmental degradation, corruption, e-government maturity, and regional grouping as variables. It is found that the maturity of e-government services could mitigate corruption leading to the economic growth of a country. In addition, Kunieda et al. (2014) demonstrated a negative impact of corruption on economic growth because highly corrupt countries impose higher tax rates than their less corrupt counterparts. Finally, Shittu et al. (2018) investigated the long-run relationship between debt, economic growth, and corruption by using data from 5 Sub-Saharan African countries in the 1990–2015 period, applying FM-OLS and dynamic OLS techniques to formulate the variables per the granger causality test. According to their findings, there is a negative relationship and bidirectional causality between debt and economic growth.

The last group of studies used models in which, depending on the intensity of the variables under consideration, the relationship between corruption and economic growth was positive, negative, or neutral. Méndez and Sepúlveda (2006) and Aidt et al. (2008) concluded that corruption has a significant negative impact on the development of countries with high-quality 'political institutions.' However, corruption does not affect development in countries with low-quality political institutions. Méon and Sekkat (2005) proposed a test for the "grease the wheels" and "sand in the wheels" hypotheses of corruption. The use of correlations between indicators that measure the quality of institutions and corruption has shown that corruption is more inhibitory in the presence of low-quality of governance and vice versa. Huang and Ho (2017) showed that Asian countries did not exhibit significant causality from governance to economic growth. Finally, the empirical findings of some recent studies, including D'Agostino et al. (2016), Sharma and Mitra (2015), and Huang (2015) for cross-country panels have yielded mixed results.

While the issues of corruption, investment, governance, growth, and development have elicited considerable attention in the extant literature, there is a paucity of research that specifically investigates the effects of corruption (Everhart et al., 2009). Many studies have focused on the causes and determinants rather than the effects of corruption (Shleifer and Vishny, 1993; Gyimah-Brempong, 2002).

From the previous literature review, it can be inferred that the simultaneous effects of various factors on the correlation between corruption and economic growth still have much to contribute. Also, although the recent literature tends towards the "sand in the wheels" hypothesis, the issue remains open with many possible extensions. Our research provides evidence of these relationships using a growth model involving governance variables (through corruption), investment, workforce quality, trade freedom, government size, innovation, and the financial sector. In addition, the direction of the correlation is tested in many models and with various econometric methods.

2.2. Transmission channels literature

As mentioned in the previous subsection, a common finding in the literature is that corruption hinders economic growth. However, there is no consensus on the mechanisms by which corruption has such an effect. In addition to the direct impacts of corruption, there are also indirect effects, referred to as transmission channels through which corruption influences growth rates. For example, corruption negatively affects investment, whose reduction can negatively influence economic growth rates (Mauro, 1995). Several authors have argued that the effects of corruption on economic growth are not adequately measured via the econometric models, particularly when the determinant equation has other variables affected by corruption (Mo, 2001; Dreher and Herzfeld, 2008). Therefore, there is a need to understand the indirect effects which include investment, human capital, political instability (Mo, 2001), free trade (Pellegrini and Gerlakh, 2004), land distribution inequalities, geographical disparities (Mishra and Ray, 2010), labour force, GDP composition (Ibrahim et al., 2015), tax evasion, and bad loans (Park, 2012).

Corruption negatively affects domestic and foreign investment (FDI) as it deters potential investors, who turn to safer havens with less uncertainty (Mauro, 1995). On the one hand, it reduces the quality of human capital as there would be a lack of jobs being offered according to knowledge and experience. On the other hand, it expels the most educated in areas that can perform better (Mo, 2001). It causes political instability because it can lead to an unjust distribution of wealth based on the existing inequalities in opportunities and disparities. This process can provoke frustration as well as socio-political instabilities. It hinders free trade, only allowing corrupt transactions to thrive and preventing free competition. Moreover, it restricts free access to work by changing the hiring processes through skills and experience. In addition, productive activities will be driven mainly by those in which corrupt transactions take place. Ultimately, there will be a reduction in tax revenues and an increase in bad loans as corruption would intervene in free taxation and allow the bad loans through procedures that go beyond the logic of financial investment (Park, 2012). Finally, it slows down innovative activities because innovators need government-provided goods (import licences) more than established producers, as the former are often limited in credit (Mo, 2001).

Two different schools of thought measure the indirect effects of corruption on economic growth through transmission channels. The first uses the decomposition methodology, in which the impact of each channel is calculated separately to compose their later effects (Mo, 2001; Pellegrini and Gerlagh, 2004). The advantage of this method is that the effects are calculated with a precise mathematical methodology. The disadvantage is that only a few channels can be estimated simultaneously, as statistical significance is required for each one. The second approach uses a system of simultaneous

equations. This methodology excludes corruption from the growth regression and estimates its effects on each of the channel variable (Dreher and Herzfeld, 2005; Dridi, 2013). Multiplying corruption's impact on a channel with the channel's effect on growth yields corruption's growth impact in a specific channel. The advantage of this method is that many channels are calculated together, while the disadvantage is that the process is entirely empirical.

Mo (2001) developed an analytical framework to examine the effects of corruption and the channels through which it impacts GDP growth. The channels under consideration included investment, human capital, and political stability. Using OLS regression analysis for 45 countries from 1970 to 1985, he identified a significant negative correlation between corruption and economic growth. However, the effect became statistically insignificant after introducing other determinants such as investment, human capital, and political stability. This finding led the author to conclude that apart from exerting a direct effect, corruption is likely to indirectly affect the growth rate through its impact on these three factors. Many researchers have conducted similar studies, introducing various transmission channels (Lambsdorff, 2003; Gründler and Potrafke, 2019).

Dridi (2013), considered investment, human capital, government spending, inflation, and political instability as transmission channels. Utilising a sample of 87 countries from 1980 to 2002 with the equation methodology, he found that corruption reduces economic growth by negatively affecting human capital and investment and positively affecting inflation and political instability. Overall, these four transmission variables lessen the growth rate by 1.36%. Finally, Gründler and Potrafke (2019) studied various channels of transmission of corruption in economic growth, including foreign direct investment, total investment, inflation, the size of the public sector, and human capital.

One of the main aims of our research is the dynamic study of the interrelationships between corruption, innovation, and economic growth. However, in the empirical literature, innovation has not been used as a channel to ascertain the ramifications of corruption for economic growth. The present research using the above methodology aims to fill this gap.

2.3. Corruption, innovation, and economic growth

Theoretical studies

The relationship between innovation and economic growth was understood and analysed in the theories of economic growth from their inception (Schumpeter, 1934; Romer, 1986). Although entrepreneurship and innovation are widely regarded as critical economic growth drivers, researchers have yet to document a strong correlation between these variables (Acs and Audretsch, 2005).

The relationship between corruption and innovation is complex and bi-directional. Innovative businesses interact with government officials and civil servants more often, so they are more prone to corruption than others. Their work requires operating licences, security inspections, and approvals from regulators. However, corruption creates a vague economic environment, resulting in innovative "financial contracts" having no fixed value. In turn, this acts as a disincentive to investment. Corruption also creates disincentives to invest in innovation by increasing transaction costs (Luhmann and Gambetta, 1988).

According to Murphy et al. (1993), the degree of protection of property rights is the most decisive factor in the distribution of talent and innovative practices. If they are not protected due to a lack of institutions, entrepreneurship becomes less attractive, and creative activity slows down. Luo (2005) investigates the effects of corruption on innovative activity in terms of organisational theory. According to him, the development of a business depends mainly on innovation. In a corrupt country, a company is based on bribery. He believes that corruption is a faster and perhaps more effective tool. These procedures are also related to the aforementioned uncertainty theory. Corruption erodes confidence in government. Innovative companies need government services to acquire new technologies as quickly as possible. Unreliable government increases uncertainty about if and how quickly appropriate approvals are provided to businesses (Anokhin and Schulze, 2009).

Kong et al. (2020) used a quasi-natural experiment to find evidence for the relationship between corruption and innovation. It investigates that anti-corruption plays an essential role in ensuring positive innovation. The researchers conducted the study in China; they calculated the total factor productivity (TFP) at the firm level under an anti-corruption campaign. Using the difference-in-differences estimation technique to calculate the TFP level, they showed that TFP increased by 1.7% due to anti-corruption measures. This research theoretically, along with calculations, proved that anti-corruption is an excellent approach to improving China's economy and innovation status.

While there is ample evidence of the negative effects of corruption on innovation, Leff (1964) and Lui (1985) argued that corruption enhances efficiency through two processes: it not only forces the government to move more quickly in terms of permissions and assistance in innovative procedures, but it also encourages competition for the limited resources available from the state. Corruption also interferes with allocating these scarce resources leading to an outcome in which the most efficient companies can pay the highest bribes. Corruption boosts innovation if the most efficient companies are also the most innovative.

Examining the relationship between corruption, transaction costs, asset specificity, and innovation through theoretical means, Barasa (2018) suggested a direct and positive link between corruption and innovation in African countries. Testing this relationship is done through a multiple mediation model, where the techniques used are bootstrapping and the coefficients approach. A positive relationship is proven between corruption and innovation, such that the business environment is affected proportionately due to the higher degree of corruption.

There is a lack of consensus in the theoretical literature on the type and sign of the relationship between corruption and innovation. Therefore, more thorough research is needed to analyse the processes through which the two variables interact in order to determine their effect on economic growth. The present study attempts to shed light on this issue.

Empirical studies

Empirical research is dominated by a general view, as also mentioned in the theoretical literature, that corruption hinders innovation (see, among others, (DiRienzo and Das, 2015; Wen et al., 2020). Specifically, DiRienzo and Das (2015) studied 113 countries in the year 2008 and found that corruption negatively impacts innovation at the country level and depends on the degree of economic development. As economic development rises, the positive impact of decreased corruption on innovation decreases. They also highlighted the significance of other factors, such as policies supporting economic freedom, development, income equality, and cultural diversity. In addition, Wen et al. (2020) examined the effect of corruption on domestic innovation using annual data for the period 1996–2013 for 29 OECD countries; they found that less corrupt countries contribute to higher innovation performance.

Pluskota (2020) found nonlinear links between corruption and economic growth and innovation (measured by the number of patent applications) across 24 developed European countries from 1996 to 2017. The dynamic GMM technique shows that increased innovation results from less corruption in the economy. She also demonstrated that corruption hinders innovation while economic growth boosts it. The effects of innovation and corruption on economic growth for 13 emerging Asian economies from 2009 to 2018 are examined by Das et al. (2020a). They demonstrated that, despite corruption, innovation has no significant impact on growth using worldwide innovation and corruption indexes. Overall, the findings show that corruption is the main barrier to long-term progress and that regional innovation is not strong enough to draw significant economic growth.

At the corporate level, studies have also been conducted in diverse countries. Mahagaonkar (2010a,b) offers analytical viewpoints on the impact of corruption on innovative activities in 3477 firms from 2002 to 2004 for various African countries. The empirical findings imply that corruption enhances marketing innovation while negatively affecting product, process, and organisational innovation. Studying U.S. companies, Ellis et al. (2020) found a robust negative correlation between corruption and innovation. Their research used long-term data from 64 countries and revealed a negative non-linear correlation between corruption and three independent business and innovation measures. Similar conclusions were reached in the study by Goel and Saunoris (2020). The previous studies highlighted the near-sighted relationship; however, a long-term relationship is also derived between corruption and innovative activity, especially in the U.S. The annual data from 48 U.S. states were collected between 1977 and 2006. The data through a panel applied the exploit time series to derive a relationship (Dincer, 2019). Two different measures were used to evaluate the innovativeness with the rate of corruption. The innovative activity was cointegrated with corruption using FM-OLS such that the results proved that corruption slows down innovative activity. Therefore, it is established that corruption acts as a decelerating factor for the growth of a country. However, as the trends vary from country to country, this notion cannot be applied to all countries. Using a sample of companies from the World Bank Business Survey for 2006–2016 in emerging and developing countries, Lee et al. (2020) found that corruption has a negative impact on innovations, mainly in the manufacturing industry.

There is also a contrasting viewpoint that corruption can lead to additional innovation. It can allow companies to influence regulatory structures in their favour, reducing the uncertainty of those involved in corruption proceedings (Nguyen et al., 2016). This process compels them to explore more innovative strategies as they operate in a "safe" environment. It has also been argued that political interconnections of companies involving "donations" reduce uncertainty and pave the way for innovation (Ovtchinnikov et al., 2020).

Islam (2021) investigated the relationship of innovation with economic growth in the Asian region by applying dynamic tools. He applied the Marshallian demand function to evaluate the dynamics of the variables. The research used the sample from 1980 to 2016 to empirically identify the dynamics using the environmental Kuznets curve (EKZ), demonstrating that the GDP increase depends on innovativeness and energy consumption. Thus, this study can be used to show the positive relationship between economic growth and innovation in the Asian region.

Regarding the impact of innovation on economic growth, the financial crisis of 2008 revealed the interdependence of these sectors of the economy. More research is needed to thoroughly analyse this interdependence. However, there has yet to be a consensus about the linkages between innovation and economic growth. Many studies find a robust empirical relationship between innovation and economic growth (Wu et al., 2016; Hasan and Tucci, 2010; Cetin, 2013; Maradana et al., 2017; Pradhan et al., 2018). Specifically, Granger causality tests were used in the work of Wu et al. (2016) and Pradhan et al. (2018) identified a positive and significant relationship between innovation and economic growth. In contrast, (Bilas et al., 2016) found a negative relationship between innovation and economic growth in their research for 28 European countries from the period 2003–2013. Finally, other researchers are yet to identify a causal relationship between the two variables (Samimi and Alerasoul, 2009; Kacprzyk and Doryń, 2017) . Research has also shown that economic growth can increase the level of innovation. Specifically, Maradana et al. (2017), using Granger causality tests for 19 European countries from 1989–2014, identified a two-way causality between innovation and per capita economic growth.

Some studies link the two variables specifically with the level of governance. Countries with poor governance appear less innovative (Hadfield, 2008) and, as a result, do not grow sufficiently. Finally, it seems that the relationship between innovation and economic growth is complex, and the country's specific characteristics play an essential role. In addition, governments can play an important role in fostering innovation (Mtar and Belazreg, 2021).

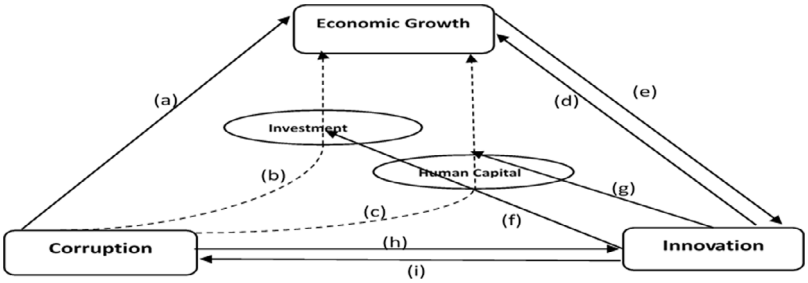

Fig. 1. The direct and indirect relationship between corruption, innovation, and economic growth.

The contribution of innovation to the relationship between the impact of corruption on economic growth has not been sufficiently explored. There is a small but growing body of literature on this issue (Ogun, 2018; Das et al., 2020b; Pluskota, 2020). Ogun (2018) observed that innovation and corruption were credible determinants of economic growth. Das et al. (2020b) found that the impact of innovation on economic growth in 13 emerging Asian economies during 2009–2018 is negligible. In addition, they have shown that corruption enhances economic growth through the presence of innovation. Pluskota (2020) investigated how corruption and economic growth affect innovation.

The nexus between corruption and economic growth was established via the influence of the former on productivity growth. However, the impact of innovation on the relationship between corruption and economic growth needs to be thoroughly studied. Fig. 1 illustrates the complex relationship between the three variables.

A vast body of literature has shown that corruption negatively affects economic growth directly (line (a) in Fig. 1) and indirectly through transmission channels (lines (b) and (c)). Innovation has a positive effect on economic growth (line (d)) (Das et al., 2020b). However, it can also indirectly affect the negative relationship between corruption and economic growth by impacting some of the transmission channels, especially the investment (line (f)) and the quality of the workforce (line (g)). When innovation is strengthened in a country, companies move to competitive models and increase their investments mainly in innovative processes. They also begin to seek more knowledgeable staff creating demand for respective jobs. This process can lead to improved human capital in an economy. Finally, growth has a positive effect on innovation (line (e)) because of the greater resources available to the economy (Schumpeter, 1934; Romer, 1986).

To summarise, both theoretical and primary empirical studies show, in general, that corruption reduces economic growth, innovation enhances growth, and innovation has a negative bidirectional relation with corruption (lines (h) and (i) in Fig. 1). However, the processes by which innovation affects the relationship between corruption and growth and how corruption affects economic growth through innovation have not been adequately studied. This study aims to fill these research gaps. In Section 4, we will explore the relationships that arise from the various lines illustrated in Fig. 1, explaining the policy implications of each.

3. Data and methodology

3.1. Data and descriptive statistics

Our sample consists of annual data covering 2010–2018 for a sample of 109 countries[1]. We used a set of macro variables: GDP per capita, General Government expenditure, Trade openness, broad money (M3/GDP), population growth, education, total investment over GDP, and the percentage of citizens who have participated in secondary education. In addition, our data set is enriched by the inclusion of innovation and corruption indices. Our innovation indices are Trademark applications, Patent applications, and R&D expenditures (% of GDP). Moreover, we used as corruption measures the Corruption Perception Index (CPI), the Control of Corruption Index (CCI), and the International Country Risk Guide Corruption index (ICRG). The period was selected to have a rich data set for all the above variables.

Per capita GDP at constant prices (LGDP)

The variable used is the GDP per capita at constant dollar prices in 2011 (in $ U.S.) in power purchasing parity. Many authors have previously used it in the same field (Lederman et al., 2005; Aixalá and Fabro, 2008; Baliamoune-Lutz and Ndikumana, 2008; Acemoglu et al., 2019; Das et al., 2020b). The data were obtained from the World Bank databaseWorld Bank (2019) by filling in local sources and updating the website (www.statista.com) for 109 countries from 2010 to 2018. Finally, the Napier logarithm of the variable was obtained.

[1] See Appendix F for the countries used in this study.

Table 1
Variables, Symbols, Period, Price Scale, Sources, and Literature.

Symbols	Variables	Time	Data Sources	Literature
LGDP	GDP per capita at fixed $ U.S. prices (2011) (PPP)	2010–2018	World Bank 2019	(Aixalá and Fabro, 2008; Acemoglu et al., 2019; Das et al., 2020b)
CPI	Corruption Perception Index	2010–2018 0–100 (0 low corruption)	Transparency International	(Pellegrini and Gerlagh, 2004; Gründler and Potrafke, 2019)
CCI	Control of Corruption Index	(−2,5 – 2,5, transformed in 0–100, rev)	World Bank	(Chen et al., 2015; Menard and Weill, 2016)
ICRG	International Country Risk Guide (Corruption Index)	2010–2018 (0 −6)	Transparency International	(Davoodi and Tanzi, 1997; Méndez and Sepúlveda, 2006)
INVT	Total Investments (% GDP)	2010–2018	World Bank, Global Econ.	(Mauro, 1995; Mo, 2001; Méon and Sekkat, 2005)
POP	Population growth (annual %)	2010–2018	World Bank	(Barro and Sala-i Martin, 2004)
SEDU	Percentage of citizens who have participated in Secondary Education	2010–2018	Global Economy	(Méndez and Sepúlveda, 2006; Chang and Hao, 2017)
GEXP	General government final consumption expenditure (% of GDP)	2010–2018	World Bank	(Easterly and Rebelo, 1993)
TROP	Total exports and imports as a percentage of a country's GDP at constant prices	2010–2018	Global Economy	(Cieślik and Goczek, 2018; Chakravorty, 2019)
BRM	Broad Money (M3/GDP)	2010–2018	W.B., IMF, OECD	(Everhart et al., 2009; Song et al., 2020)
RDEXP	Research and development expenditure (% of GDP)	2010–2018	WB	(Blackburn and Powell, 2011; Das et al., 2020b)
TRMRK	Trademark applications, total (per capita, per million)	2010–2018	WB	(Grupp and Schubert, 2010; Wen et al., 2018)
PAT	Patent applications, all (per capita, per million)	2010–2018	WB	(Ellis et al., 2020; Dincer, 2019)

Corruption indicators (CPI, CCI, ICRG)

Subjective methods of measuring corruption will be used (measuring perceptions) in this study: Transparency International (2018) created CPI (aggregate index) ranks countries on a scale from 0 (high corruption) to 100 (low corruption). Due to the post-crisis analysis and data reliability, we used corruption data for 109 countries (period 2010–2018). We have reversed the index so that 0 means minor corruption and 100 means large corruption for a better display and discussion of the results.

The CCI is one of the six dimensions of the World Bank's world governance indicator (WGI); it compiles governance indicators, and its structure is similar to that of the CPI, which it has sought to improve in some areas (Kaufmann et al., 1999). The measurement of corruption with the Control of Corruption Index is one of the index dimensions. Corruption levels are measured on a scale from −2.5 to 2.5, with 2.5 as the best level of non-corruption. It was taken for 109 countries from 2010 to 2018 before being converted to a 100-degree scale and reversed.

Finally, the ICRG indicator is used, and the rating is 6 points. Published by the PRS Group (TABLE 3B: Researcher's Dataset: Corruption), it has been calculated for 141 countries from 1984 to 2018. It will be used for 109 countries from 2010 to 2018. Table 1 presents the variables, the number of countries, and the existing literature.

Innovation indicators (RDEXP, TRMRK, PAT)

In this research, three measurement indicators are established: the primary innovation indicator is the R&D expenses functioning as a percentage of a country's GDP. The index is reliable, as it also measures the imported investments in innovations. It is taken from the WB database research and development expenditure (% of GDP) data (https://data.worldbank.org/indicator/GB.XPD.RSDV.GD.ZS). It will be used for 109 countries from 2010–2018.

The other two indexes used are patents and trademarks. Both figures are taken per million inhabitants to be measured and compared in countries with low innovation activity (Grupp and Schubert, 2010). There are measurements for residents in a country, non-residents, and the total. Total sizes were used in this study. Patents are a qualitative measure of innovation, as patent procedures are complex and time-consuming (Ellis et al. 2019; Dincer, 2019).

Finally, a trademark is one variable that has been used in recent years. According to the World Intellectual Property Organisation (WIPO, 2004), a trademark is "a distinctive mark that identifies certain goods or services such as those provided by a particular person or business". This quantitative variable contains a broader innovation concept and is often overestimated (Mendonça et al., 2004; Grupp and Schubert, 2010; Wen et al., 2018).

3.2. The empirical model

The equation of the complete integrated OLS system (FM-OLS) is similar to what was proposed by Gründler and Potrafke (2019) and Everhart et al. (2009) as follows:

$$y_{\mathrm{it}} = \beta_0 + \beta_1 \mathrm{C}_{\mathrm{it}} + \sum_{j=2}^{m} \beta_j x_{it}^{j} + \sum_{r=1}^{d} a_r z_{it}^{r} + \eta_{\mathrm{t}} + \mu_{\mathrm{i}} + \upsilon_{\mathrm{it,}} \tag{1}$$

where y_{it} denotes GDP per capita (natural logarithm), C_{it} is corruption and x_{it}^{j} are m-1 explanatory variables, which can indirectly change the GDP per capita (transmission channels). These are investments in private, public, and human capital. The percentage of the population participating in secondary education (quality of the workforce) and innovation. z_{it}^{r} are the other independent variables that affect per capita GDP without being particularly associated with corruption. These include trade openness, population growth rate—a proxy for labour, broad money, and the size of public sector-government expenditure. μ_{i} refers to the constant individual effect of each country's attributes, η_{t} signifies the time effects (shocks such as natural disasters, wars, economic or other crises, etc.), and υ_{it} represents the time-dependent error.

More specific equations will be used for studying the effect of combined factors such as corruption and innovation (C_{it} X x_{it}, cross term in the equation, C_{it} as corruption indicator, and x_{it} as innovation indicator).

Finally, the direct effects of corruption on per capita GDP will be combined with indirect effects.

3.3. Econometric methodologies

OLS, PCSE, and PCSE(AR1)

First, we examine whether there is a serial correlation using the Durbin–Watson test for OLS econometrics. Next, we proceed to the Breusch–Pagan test for heteroscedasticity (Koenker and Bassett Jr., 1982). We use the White estimators to correct the prevailing heteroscedasticity. However, a study of the variables and correlations of our research necessitates more dynamic data because our results must take into account both time and differences between countries. This leads us to the panel data analysis.

The probability of the omitted variables, the heterogeneity, and the multicollinearity issues prompt us to perform various tests such as the t-test, Breusch–Pagan/Cook–Weisberg test, and Ramsey and F-test. The Breusch–Pagan/Cook–Weisberg test revealed a possible problem of heterogeneity in the country data. Hence, OLS methodologies that address this particular issue were initially used. Applying the Wooldridge test for autocorrelation, modified Wald method, and Breusch Pagan test for heteroscedasticity to identify the problems of autocorrelation and heteroscedasticity, we introduce PCSE models with panel-corrected standard errors [PCSE and PCSE (AR1)] (Cameron and Trivedi, 2009).

Fully Modified OLS dynamic methodology

We use the FM-OLS method to deal with endogeneity issues and the problems of omitted variables. Initially, we examine whether the variables in the model are stationary. Since the variables under investigation are cointegrated, we can estimate their relationship in the long-run equilibrium state. When applied to cointegrated panels, the least-squares method (OLS) leads to discriminatory estimators, especially when the sample size relative to periods (t) is small. Therefore, we estimate the long-run equilibrium function with the FM-OLS procedure proposed by Pedroni (2001) to calculate valid t-statistics. This approach yields more reliable results when heterogeneity in integrated variables is diagnosed. The primary equation used in the FM-OLS model is (1). The constant β_0 includes the heterogeneity of each country, which may differ. C_{it}, $x_{\mathrm{it,}}$ and z_{it} variables are first-order integrated, and the error term tends asymptotically to the normal distribution.

Finally, several economic growth authors have used these methodologies (see, among others, Dincer, 2019; Song et al., 2020).

Dumitrescu–Hurlin causality test for panel data

In Appendix A, we present the methodology proposed by Dumitrescu and Hurlin (2012) and use it to find a causal relationship between the variables.

4. Empirical results

In Appendix B, we present the statistical characteristics of the variables (Table B.1).

4.1. Preliminary tests of econometric analysis (FM-OLS methodology)

Out analysis entails using a multivariate panel model, referring to 109 countries (Appendix F). As mentioned above, the data set includes annual observations for the following variables: logarithm per capita GDP (LGDP), corruption (CPI, CCI, ICRG), total investment in GDP units (INVT), annual average population growth (POP), and participation in secondary education (SEDU). Additionally, imports and exports in GDP units (TROPEN), capital circulation in GDP units (BRMON), government expenditure in GDP units (GEXP), and innovation (RDEXP, PAT, TRMRK) are used as well. To begin with, we performed stationary tests of all the variables in the sample.

We use unit root tests introduced by Levin, Lin and Chu, Breitung, Maddala and Wu, and Choi. Table C.1 in the Appendix B shows the results, including values at the levels and the first differences. According to the findings, all our variables are stationary in the first difference. The null hypothesis of nonstationarity is rejected at the 99% significance level for most tests.

Cross-sectional dependence (CSD) and panel cointegration results.

The unit root tests assume that the cross-sectional units of the data panels are not strongly correlated. Next, we run the CSD tests proposed by Pesaran and Frees with a model containing nine variables. Its basic form, the CCI indicator, is used as corruption and R&D expenses as an innovation indicator. The other variables of corruption (CPI, ICRG) and innovation (TRMRK, PAT) will be utilised to ascertain the robustness of the basic model.

Cross-sectional dependence is also confirmed in the models, thus implying that one change in one country is likely to affect others. This process occurs as similar processes between neighbouring countries across several areas such as tourism, free trade rules, legislation, governance, corruption, etc. (Gründler and Potrafke, 2019). A typical example is the European Union, NAFTA, Latin countries, South Africa, etc. To handle and mitigate the issue of cross-sectional dependence, the PCSE methodology is used (Hoechle, 2007) in addition to FM-OLS econometrics (Baltagi, 2006; Costantini and Destefanis, 2009).

To apply FM-OLS econometrics, we adopt panel cointegration tests proposed by Kao, Pedroni, and Westerlund to determine the possible existence of a cointegrated vector in our essential variables. The results show cointegration between the model's main variables, LGDP as a dependent variable, CCI, POP, SEDU, INVT, and RDEXP as regressors, and a long-term equilibrium relationship.

4.2. Empirical results of the impact of corruption and innovation on GDPpc

In the present subsection, we explore the effects of corruption and innovation on economic growth, where the effects of lines a and d of Fig. 1 will be clarified. The previous section presented Eq. (1) of the basic model. The proposed analysis includes initial correlations with fixed and random effects display of Hausman, Woolridge, Wald, Breusch–Pagan, Pesaran, and Frees tests, where the model seems to have heteroscedasticity, autocorrelation, and cross-sectoral dependence. As shown in Table 2, the PSCE and PSCE (AR1) models are used for correction to consider the past effects. Finally, as the endogenous problem arises, we end up with the fully modified OLS methodology, which considers the past temporal effects on the variables.

Key conclusions of the empirical model

Tables 2 and 3 exhibit a strong negative relationship between corruption and economic growth in each sample and any econometric methodology (line a of Fig. 1). Table 2 shows that this relationship is not impacted by the method of estimation even when an autoregressive factor is included in the model, increasing its explanatory power. Focusing on Table 3, we argue that in the sample and with FM-OLS, the increase of corruption by one unit leads to a reduction in GDP per capita by 1.05%, a trend that remains strong in all models with CCI, CPI, and ICRG corruption as well as with RDEXP, TRMRK, and PAT innovation indicators. This percentage effect ranges from 1.05% to 1.55%. These conclusions are consistent with the literature and the "sand in the wheels" hypothesis (Mauro, 1995; Pellegrini and Gerlagh, 2004; Cieślik and Goczek, 2018).

Concerning the rest of the explanatory variables and in line with the literature, the increase of POP by one unit leads to a decline of the GDPpc by 1.3% due to the decreasing returns scale of labour (Barro and Sala-i Martin, 2004). Furthermore, the growth of the public sector (GEXP) by one unit of GDP lowers GDPpc by 2.9%. Finally, it can be observed that an improvement in innovation leads to an enhancement of GDPpc (line d of Fig. 1). Moreover, an increase in money invested in research and development significantly improves GDPpc (Blackburn and Powell, 2011; Das et al., 2020b). Increasing trademarks by 1 in 1,000 people also boosts GDPpc (Grupp and Schubert, 2010; Wen et al., 2018). However, patent improvement generally does not have a statistically significant effect.

Robustness checks

(A) Econometric methodologies and endogeneity

Table 2 presents the models through various econometric methods, giving similar estimators with stable signs to our basic variables (rows 1, 2, and 5). After identifying the cointegration and the cross-sectional dependence between these variables, we utilised simple and dynamic error correction methods and the FM-OLS methodology.

(B) Different variable models

In rows 3, 4, and 5, we used gradual completion of our final model. First, we used our two basic variables (row 3). We initially added the key variables used by the literature in growth models: physical capital, labour, and human capital quality in row 4 (Mauro, 1995; Méon and Sekkat, 2005; Aixalá and Fabro, 2008; Anoruo and Braha, 2005). Finally, we introduced our full empirical model derived from Eq. (1). We find stability in the standard variables and a proportional increase in the models' completeness as R^2 gradually increases.

(C) Sensitivity test to alternative indicators

In Table 3, we use three different variables to measure corruption the CPI (rows 1–3), the CCI (rows 4–6), and the ICRG (rows 7–9) to observe stability in the effect of corruption on economic growth from −1.05% to −1.55%. Additionally, we

Table 2
Main results with error correction and FM-OLS estimation method.

Dependent variable	Logarithm of GDP per capita: LGDP981 observations in 109 countries (total)				
Model	PCSE	PCSE (AR1)	FM-OLS (1)	FM-OLS (2)	FM-OLS (3)
CCI	−0.012140*** (0.000)	−0.015190*** (0.000)	−0.040582*** (0.000)	−0.018088*** (0.000)	−0.010485*** (0.000)
POP	−0.091584*** (0.000)	−0.054650*** (0.006)		−0.094844** (0.023)	−0.129956*** (0.000)
SEDU	0.020819*** (0.000)	0.014533*** (0.000)		0.021695*** (0.000)	0.019028*** (0.000)
INVT	0.000254 (0.881)	0.001177 (0.403)		0.021598*** (0.000)	0.013518*** (0.000)
TROP	0.002066*** (0.000)	0.001621*** (0.000)			0.002186*** (0.000)
BRM	0.001128*** (0.000)	0.001605*** (0.000)			0.001282*** (0.000)
GEXP	−0.024335*** (0.000)	−0.004045 (0.247)			−0.029479*** (0.000)
RDEXP	0.179783*** (0.000)	0.159361*** (0.000)			0.196636*** (0.000)
R^2	0.8188	0.9879	0.1898	0.2823	0.4250
Hausman Test	Chi2(8) = 12.15 (0.1446)				
Wooldridge Test for autocorrelation	F(1,108)= 948.432***				
Modified Wald Test for group-wise heteroscedasticity	Chi2(109) = 20777.64***				
Breusch Pagan Test heteroscedasticity	LM=3222.46*** (0.0000)				
		Rho(ar1) = 0.891676			

*, **, and *** mean the significance level of 10%, 5%, and 1%, respectively, the numbers in parentheses show the p values. Variables and sources are as same as in Table 1 with the author's calculations. In FM-OLS, lags 2.00 are used, and the linear and quadratic eqtrend effect is controlled (2).

use three different variables to measure innovation R&D expenses (rows 1, 4, 7), trademarks (rows 2, 5, 8), and patents (rows 3, 6, 9). There is a consistency in the positive impact of innovation on economic growth in all models.

(D) Subsamples due to developmental classification

Table 4 displays the correlation of the primary growth variables with the addition of corruption and innovation with GDP (models 1, 3) and the full models (2 and 4) in two subsamples. To separate countries according to development performance, the classification of the World Bank was used where the criterion is per capita income (GNI). This separation aims to control the consistency of the sample, and we want to investigate whether our basic variables have different effects in countries with varying developmental levels. The first subsample comprises low and lower-middle-income countries, whereas the second consists of high and upper-middle-income countries. Corruption in all models confirms the vital negative sign and its harmful impact on economic growth, while innovation has a positive relationship in the subsamples. Moreover, coherence is shown in all variables.

The results show that the variables' signs and significance are constant, and only minor changes are made. These results are robust to the changes in the measurement scales and the constraints of the models.

(E) Regional subsamples according to the World Bank's classification. Static and dynamic panel regression models

Table 5 displays the correlation of the growth variables with the addition of corruption and innovation with GDP in five subsamples. The World Bank's classification was proposed to make this division, such that the criterion is the region to which the country belongs (Fethi and Imamoglu, 2021). This separation aims to reduce the estimation bias and investigate the essential factors affecting the growth in each region. The first region comprises 40 Europe and Central Asia countries, and the second consists of 17 Latin American and Caribbean countries. The third region consists of 14 East Asia and Pacific countries, the fourth of 9 Middle East and North African countries, and the fifth of 24 Sub-Saharan and African countries. However, no analysis was conducted in North America and South Asia due to the few countries included in the model. Corruption confirms the vital negative sign and its harmful impacts on economic growth outside of MENA and SSA areas where it boosts economic growth. Innovation has a positive relationship in all the subsamples. In the econometric analysis, two methods are used—a static one, that is, the PCSE, and a dynamic one, or the FM-OLS, to avoid any inconsistencies.

Moreover, coherence is indicated in all variables. We observe an agreement in the directions of the variables between the static and dynamic methods. Nevertheless, in general, there are also significant differences between the regions. In ECA countries, corruption hinders economic growth. However, it is not a critical factor (0.4% decrease in GDPpc with 1% increase in corruption), such as population growth, investment, and innovation, which positively affect economic growth.

Table 3
Effects of corruption and innovation on growth with FM-OLS estimation method, using CCI, CPI, ICRG Corruption and RDEXP, TRMRK, PAT Innovation indicators.

Dependent variable	Logarithm of GDP per capita: LGDP981 observations 109 Countries								
Model	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
CCI	−0.010485*** (0.000)	−0.015452*** (0.000)	−0.013894*** (0.000)						
CPI				−0.012237*** (0.000)	−0.015020*** (0.000)	−0.011923*** (0.000)			
ICRG							−0.142987*** (0.000)	−0.228860*** (0.000)	−0.202202*** (0.000)
POP	−0.129956*** (0.000)	−0.151643*** (0.000)	−0.192542*** (0.000)	−0.100526*** (0.000)	−0.138702*** (0.000)	−0.160503*** (0.000)	−0.101482*** (0.000)	−0.124088*** (0.000)	−0.112034*** (0.003)
SEDU	0.019028*** (0.000)	0.019369*** (0.000)	0.020003*** (0.000)	0.020410*** (0.000)	0.019956*** (0.000)	0.021511*** (0.000)	0.022253*** (0.000)	0.022252*** (0.000)	0.023985*** (0.000)
INVT	0.013518*** (0.000)	0.013121*** (0.000)	0.015034*** (0.010)	0.010721 (0.000)	0.013527*** (0.000)	0.015310*** (0.009)	0.011841*** (0.000)	0.013756*** (0.000)	0.013368** (0.016)
TROP	0.002186*** (0.000)	0.000550** (0.043)	0.001694** (0.019)	0.001711*** (0.000)	0.000735** (0.034)	0.001450** (0.046)	0.002787*** (0.000)	0.001075*** (0.005)	0.001698** (0.012)
BRM	0.001282*** (0.000)	0.000254 (0.503)	0.001051 (0.128)	0.001678*** (0.000)	0.000434 (0.216)	0.001544** (0.026)	0.000879*** (0.000)	0.000517 (0.184)	0.001014 (0.122)
GEXP	−0.029479*** (0.000)	−0.020685*** (0.000)	−0.023249*** (0.000)	−0.031326*** (0.000)	−0.022852*** (0.000)	−0.021632*** (0.001)	−0.033983*** (0.000)	−0.027903*** (0.000)	−0.022948*** (0.000)
RDEXP	0.196636*** (0.000)			0.183712*** (0.000)			0.208972*** (0.000)		
TRMRK		0.0000897*** (0.000)			0.0001054*** (0.000)			0.0000931*** (0.000)	
PAT			0.0000652 (0.391)			0.0000657 (0.391)			0.0001329* (0.061)
R^2	0.4250	0.5327	0.2758	0.5233	0.4806	0.3196	0.3387	0.5224	0.3061

*, **, and *** mean the significance level of 10%, 5%, and 1%, respectively, the numbers in parentheses show the p values. Al abbreviations and sources are the same as in Table 1 with the authors' calculations. In FM-OLS, lags 2.00 are used, and the linear and quadratic eqtrend effect is controlled (2).

Table 4
Long-term effects of the FM-OLS estimation method in the low and high development countries.

Dependent variable	Logarithm of GDP per capita: LGDP981 observations in 109 countries (total)			
Subsamples	Subsample 1 — 314 obs. 35 countries		Subsample 2 — 666 obs. 74 countries	
Models	Model 1	Model 2	Model 3	Model 4
CCI	−0.013882*** (0.000)	−0.005233*** (0.008)	−0.017925*** (0.000)	−0.015953*** (0.000)
POP	−0.273014*** (0.000)	−0.088152*** (0.001)	−0.039879 (0.147)	−0.020078 (0.556)
SEDU	0.015682*** (0.000)	0.022524*** (0.000)	0.003431* (0.064)	0.002996 (0.194)
INVT	0.003931 (0.316)	0.008093*** (0.000)	0.009917* (0.066)	0.015156** (0.025)
TROP		−0.001191* (0.086)		0.002522*** (0.000)
BRM		0.004773*** (0.000)		−0.000092 (0.863)
GEXP		−0.028399*** (0.000)		−0.019710** (0.044)
RDEXP	0.579687*** (0.000)	0.260881*** (0.000)	0.149026*** (0.000)	0.21319*** (0.000)
R^2	0.3957	0.2823	0.1934	0.3205

*, **, and *** mean the significance level of 10%, 5%, and 1%, respectively, the numbers in parentheses show the p values. Al abbreviations and sources are the same as in Table 1 with the authors' calculations. In FM-OLS, lags 2.00 are used, and the linear and quadratic eqtrend effect is controlled (2).

The results can be interpreted in favour of advanced economies capable of tackling the issue of corruption compared with weaker economies as they have more efficient legal systems, better policies and economic stability, and better governance, public services, and infrastructure. These estimations are in line with those by Lau et al. (2015) and Fethi and Imamoglu (2021). Consequently, there is a need for labour in ECA countries due to the low birth rates and the economic growth following the financial crisis (an increase of 5.7% of GDPpc with a population rise of 1%). Finally, these countries' stable structures take full advantage of their innovative activities and contributions.

LAC and EAP countries show many similarities in their economic growth models. Corruption is a strong deterrent to growth (3.2% and 1.85% decrease in GDPpc, respectively, for a 1% increase in corruption), and investment remains a potent stimulus. LAC countries are experiencing large-scale corruption due to political instability that undermines economic performance, while investment can lead to growth. In addition, population growth and government spending reduce economic growth as these economies are underdeveloped, and there is a shortage of new jobs, which ultimately may lead to immigration. These findings align with Gaviria (2002) for the LAC region and contrast with Huang (2016), who found mixed results in Asia and the Pacific countries.

In the MENA region, corruption hinders (with a 1.2% decrease in GDPpc for a 1% increase in corruption), and innovation enhances economic growth (in line with Guetat, 2006; Hakimi and Hamdi, 2017). Other vital factors are population growth (5.6% increase in GDPpc with 1% increase in population) and government size (negative). The empirical analysis revealed that the impact of institutional variables is vital in the MENA region. The same applies to the indirect effects of corruption on growth through investment and human capital. Thus, better-performing institutions are likely to improve growth by increasing the efficiency of investment and human capital. These institutions are essential for growth and productivity because they influence the incentives of growth performance mainly through cost-effective investments and the increase of innovation.

In SSA countries, corruption fosters economic growth in the static model and has a positive yet insignificant effect in the dynamic model. These results are in line with Shittu et al. (2018). In addition, innovation is a significant driver of growth in these countries (Oluwatobi et al., 2020), as are other critical positive factors, such as investment, education, and trade openness. As the structures in these countries are weak, corruption accelerates growth by overcoming bureaucratic and other problems. This shows that weak institutions, a weak rule of law, internal conflicts, high debt, poor regulation, and economic and political instability have aggravated such countries' economic growth. At the same time, the workforce impacts economic growth only because of its low cost, not because of its quality. These countries need structural reforms and political stability with processes that could attract financial aid and investment.

Results of the Granger causality test

We performed a causality test to explore the causal relationship between the variables of corruption and innovation with economic growth. Table 6 shows the corresponding values for the variables of interest:

As exhibited in lines 1–4, a bi-directional causal relationship between corruption and economic growth was expected in consonance with the theory (line a of Fig. 1). This is the case with both the CCI and CPI corruption indicators. When

Table 5
Effects of corruption and innovation on growth with PCSE (static) and FM-OLS (dynamic) estimation method, using CCI Corruption and RDEXP Innovation indicators in five regions according to the World Bank's classification.

Dependent Variable	Logarithm of GDP per capita: LGDP 981 observations 109 Countries									
Regions, Count Observations	ECA, 360 obs, 40 countries Europe and Central Asia		LAC, 153 obs, 17 countries Latin American and Caribbean		EAP, 126 obs, 14 countries East Asia and the Pacific		MENA, 81 obs, 9 countries Middle East and North Africa		SSA, 216 obs, 24 countries Sub Saharian Africa	
Model	Model 1 PCSE	Model 2 FM-OLS	Model 3 PCSE	Model 4 FM-OLS	Model 5 PCSE	Model 6 FM-OLS	Model 7 PCSE	Model 8 FM-OLS	Model 9 PCSE	Model 10 FM-OLS
CCI	−0.015435*** (0.000)	−0.004099** (0.014)	−0.010229*** (0.000)	−0.03245*** (0.010)	−0.018413*** (0.000)	−0.018485*** (0.010)	−0.023821*** (0.000)	−0.012156*** (0.000)	0.020414*** (0.000)	0.004439 (0.484)
POP	0.124868*** (0.000)	0.056823** (0.018)	−0.032759 (0.278)	−0.124337*** (0.002)	−0.033342 (0.239)	−0.042353* (0.099)	0.042896*** (0.010)	0.056237*** (0.000)	−0.376612*** (0.010)	−0.433331*** (0.000)
SEDU	0.003738*** (0.000)	0.004779*** (0.000)	0.010247*** (0.000)	0.009986*** (0.000)	0.006629*** (0.000)	0.009785*** (0.000)	0.027398*** (0.000)	0.009002*** (0.000)	0.013802*** (0.000)	0.034205*** (0.000)
INVT	0.006956* (0.057)	0.014427*** (0.001)	0.020173*** (0.000)	0.029320*** (0.000)	0.002279 (0.217)	0.019627*** (0.000)	0.027370*** (0.000)	0.006740*** (0.000)	0.000383 (0.918)	0.042786*** (0.000)
TROP	0.000651*** (0.000)	0.001112*** (0.002)	−0.002538*** (0.001)	−0.001379* (0.055)	0.002421*** (0.000)	0.002224*** (0.000)	0.003490*** (0.000)	0.007298*** (0.000)	0.001976 (0.167)	0.020907*** (0.000)
BRM	0.000713*** (0.000)	0.001112*** (0.000)	−0.000889 (0.485)	−0.004157*** (0.000)	0.000115 (0.599)	0.000639*** (0.004)	−0.005692*** (0.000)	−0.003748*** (0.000)	−0.008604*** (0.000)	−0.025397*** (0.000)
GEXP	−0.008357*** (0.000)	−0.023932*** (0.000)	−0.097331*** (0.000)	−0.136797*** (0.000)	−0.010932** (0.027)	−0.005791 (0.166)	−0.074350*** (0.000)	−0.032348*** (0.000)	−0.012815*** (0.000)	−0.023584*** (0.000)
RDEXP	0.096101*** (0.000)	0.379391*** (0.000)	0.843778*** (0.000)	0.732643*** (0.000)	0.224282*** (0.000)	0.222304*** (0.000)	0.124204** (0.038)	0.425678*** (0.000)	1.131927*** (0.000)	2.438727*** (0.000)
R^2	0.7731	0.2330	0.7581	0.4513	0.9593	0.8321	0.9320	0.9353	0.5400	0.2086

*, **, and *** mean the significance level of 10%, 5%, and 1%, respectively, the numbers in parentheses show the p values. Al abbreviations and sources are the same as in Table 1 with the authors' calculations. In FM-OLS, lags 2.00 are used, and the linear and quadratic eqtrend effect is controlled (2).

Table 6
Granger Causality tests as Dumitrescu and Hurlin (2012) performed for panel data.

α/α)	Null Hypothesis (H_0)	W-bar statistics	Z-bar statistics	p-value
1	CCI does not Granger-cause LGDP	4.2583	24.0544***	0.0000
2	LGDP does not Granger-cause CCI	3.3519	17.3627***	0.0000
3	CPI does not Granger-cause LGDP	2.6160	11.9297***	0.0000
4	LGDP does not Granger-cause CCI	2.1684	8.6259***	0.0000
5	CCI does not Granger-cause SEDU	2.5897	11.7356***	0.0000
6	CCI does not Granger-cause INVT	2.8254	13.4761***	0.0000
7	CCI does not Granger-cause RDEXP	2.2806	5.9380***	0.0000
8	CCI does not Granger-cause TROPEN	3.4397	18.0111***	0.0000
9	SEDU does not Granger-cause LGDP	3.2913	16.9155***	0.0000
10	INVT does not Granger-cause LGDP	3.0318	14.9993***	0.0000
11	RDEXP does not Granger-cause LGDP	4.4456	25.4368***	0.0000
12	TROPEN does not Gr.-cause LGDP	3.4890	18.3747***	0.0000
13	CCI does not Granger-cause PAT	2.5407	11.3743***	0.0000
14	PAT does not Granger-cause CCI	1.8396	6.1980***	0.0000
15	CCI does not Granger-cause TRMRK	2.4078	10.3927***	0.0000
16	TRMRK does not Granger-cause CCI	2.2319	9.0946***	0.0000
17	RDEXP does not Granger-cause CCI	3.1741	16.0501***	0.0000
18	TRMRK does not Granger-cause LGDP	7.3879	47.1578***	0.0000
19	LGDP does not Granger-cause TRMRK	8.7591	47.5807***	0.0000

*** means the significance level of 1%, the numbers in parentheses show the p values. Al abbreviations and sources are the same as in Table 1 with the authors' calculations. We used the test with lag=1.

corruption increases, the economy's resources can be seen to shift to less productive sectors and are distributed based on the extra earnings rather than the rule of economies.

Additionally, corruption has been found to have a causal relationship with education, overall investment, innovation, and trade openness, namely "transmission channels" of corruption on economic growth (ratios 5–12, lines b, c, h of Fig. 1).

The primary purpose of our research is to determine the effect of innovation on the corruption-economic growth correlation. To this end, we conducted a causal analysis between these variables. The results (rows 13–19) found a causal bi-directional relationship between corruption and innovation with different variable measurements (lines h and i of Fig. 1). This relationship was expected, as the main consequences of corruption include reduced innovation (some entrepreneurs transfer resources from innovation to corruption because it is more profitable for them in the short run). At the same time, increasing innovation hurts corruption, as it is premised on a fair distribution of resources and an essentially innovative process. Therefore, it does not need corrupt transactions to pay off.

4.3. Empirical results of transmission channels

Several authors have opined that the effect of corruption on economic growth is not accurately calculated in econometric models when the determinant equation includes variables affected by corruption (Mo, 2001; Dreher and Herzfeld, 2008; Gründler and Potrafke, 2019). A common finding in the previous studies has been that the relationship between corruption and economic growth significantly decreases by including other key variables. This shows that the impacts of significant corruption on growth are transmitted indirectly through the main determinants of economic growth due to corruption's negative effect on several factors that contribute to economic growth (lines b, c, h→d of Fig. 1).

Research framework of transmission channels

First, we assess the direct impact of corruption on economic growth through the growth model. Second, we calculate the effects of each channel, individually and then cumulatively, on economic growth. Third, the model estimates the impact of corruption on each channel separately. Finally, we assess whether the channels interpret a significant percentage of the indirect effect of corruption on economic growth as well as the numerical values of these effects on the overall model.

Direct effects of corruption and the channels on economic growth

We focus on the effects of corruption and the channels through which it affects GDP growth. In this regard, the channels under consideration are as follows: the investment channel (INVT) (Mauro, 1995), the human capital channel (SEDU) (Murphy et al., 1993), and the innovation channel (RDEXP) (Hasan and Tucci, 2010). We also examined the trade openness channel with a small but statistically significant result (TROPEN) and used the model of 109 countries with these variables: GDPpc, corruption (CCI), population growth (POP), broad money (BRM), trade openness (TROPEN), government spending (GEXP) and the aforementioned "transmission channels".

Table 7 reveals that the effect of corruption on economic growth decreases as the indirect variables under investigation are introduced individually (from −3.24% to −1.05%, progressively; models 1–4). All the results are found to be statistically significant.

This gradual reduction of the effect of corruption on economic growth is assumed to be due to the indirect effects from other transmission channels and is consistent with the literature (Mo, 2001; Dridi, 2013). In particular, corruption

Table 7
The effect of the transmission channels; FM-OLS estimation method.

Dep. Var.	LGDP, 981 observations in 109 countries (total)					
Model	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
CCI	−0.032360*** (0.000)	−0.030850*** (0.000)	−0.017904*** (0.000)	−0.010485*** (0.000)	−0.016625*** (0.000)	−0.020245*** (0.000)
POP	−0.416995*** (0.000)	−0.410469*** (0.000)	−0.133332*** (0.000)	−0.129956*** (0.000)	−0.094611*** (0.000)	−0.358507*** (0.000)
INVT		0.023657*** (0.000)	0.013085*** (0.000)	0.013518*** (0.000)		
SEDU			0.019994*** (0.000)	0.019028*** (0.000)	0.021421*** (0.000)	
RDEXP				0.196636*** (0.000)		0.256387*** (0.000)
TROP	0.000641 (0.438)	0.000173 (0.835)	0.001273*** (0.000)	0.002186*** (0.000)	0.001654*** (0.000)	0.001995*** (0.005)
GEXP	−0.023370*** (0.002)	−0.016427** (0.029)	−0.024892*** (0.000)	−0.029479*** (0.000)	−0.025878*** (0.000)	−0.024026*** (0.000)
BRM	0.001248 (0.104)	0.001696** (0.028)	0.001410*** (0.000)	0.001282*** (0.000)	0.001312*** (0.000)	0.001477** (0.020)
R^2	0.4454	0.2314	0.5964	0.4250	0.4891	0.2857

*, **, and *** mean the significance level of 10%, 5%, and 1%, respectively, the numbers in parentheses show the p values. Al abbreviations and sources are the same as in Table 1 with the authors' calculations. In FM-OLS, lags 2.00 are used, and the linear and quadratic eqtrend effect is controlled (2).

negatively affects investment as domestic and foreign investors are reluctant to invest in areas with high levels of corruption. They prefer areas where their investment will have less risk and uncertainty. This is shown in Table 8, where a 1% increase in corruption results in a 0.23% decrease in investment. Thus, part of the negative effect of corruption is transmitted to economic growth via the indirect impact of investment (from 3.24% to 3.08%). In addition, corruption can divert businesses and talented people from productive activities, resulting in the inefficient use of human capital and the prevention of innovative investments, which, in turn, harm economic growth. These effects are calculated through the secondary education participation and innovation channels. As shown in Table 8, corruption negatively affects SEDU and innovation (an increase of 1% results in a decrease of 0.70% and 0.03%, respectively). Thus, part of the negative effect of corruption is transferred to economic growth through the indirect impact of SEDU (from 3.08% to 1.79%) and innovation (from 1.79% to 1.05%).

Additionally, Table 7 and models 2, 5, and 6 show the individual effects on the economic growth of each transmission channel separately.

The effects of corruption on the channels

We differentiate Eq. (1) with the aforementioned variables. Then, using the derivatives, we determine the indirect effect of each channel. The overall impact of corruption on economic growth can be decomposed into a direct result and a set of indirect effects based on the following relationship:

$$\frac{dLGDP}{dCCI} = \frac{\partial LGDP}{\partial CCI} + \sum_{x} \left(\frac{\partial GDP}{\partial x} \frac{\partial x}{\partial CCI} \right),$$

x_{it}^{j} are the transmission variables. The first term of the right part of the equation is derived from the direct effect of corruption on GDPpc in the channel. The sum's first term is derived from the effect of individual channels on GDPpc, whereas the second term is derived from the effect of corruption on the GDP channel. For the last term, we used the following equations:

$$\mathrm{x_{it}} = \gamma \mathrm{C_{it}} + \alpha \mathrm{POP_{it}} + \beta_1 \mathrm{TROP_{it}} + \beta_2 \mathrm{GEXP_{it}} + \beta_3 \mathrm{BRM_{it}} + \eta_\mathrm{i} + \zeta_\mathrm{t} + \varepsilon_\mathrm{it} \tag{2}$$

We separately assess the impact of corruption on each transmission variable. Multiplying corruption's effects on the channel with the channel's effects on growth allowed us to calculate the effect of corruption on economic growth through this channel. This decomposition method showed that the growth rate is reduced by 3.24% by improving 1% in the corruption index.

Table 8 shows the effect of corruption on each channel, with the dependent variable being each channel on a case-by-case basis. In Channel 1, INVT is the dependent variable. We observe that an increase of CCI by 1% causes a decrease in total investments by 0.23% of GDP. It also causes a deterioration in the quality of human capital by 0.69% and innovation expenditures by 0.034% of GDP (Channel 3). Channel 2 seems to be the strongest based on the R2 measure.

Channels interpretation and direct and indirect impact of corruption on economic growth

Table 8
The effect of corruption on the transmission variables; FM-OLS estimation method.

Models Dependent variable	Channel 1 INVT	Channel 2 SEDU	Channel 3 RDEXP
CCI	−0.23514** (0.031)	−0.698978*** (0.000)	−0.034018*** (0.000)
POP	1.232045*** (0.000)	−12.79042*** (0.000)	−0.120832** (0.011)
TROPEN	−0.000932 (0.902)	−0.031136** (0.094)	−0.003063*** (0.004)
GEXP	0.042353 (0.536)	0.264359 (0.115)	0.029001*** (0.002)
BRM	0.004131 (0.557)	0.004431 (0.797)	0.000273 (0.780)
R^2	0.1492	0.4066	0.1522

*, **, and *** mean the significance level of 10%, 5%, and 1%, respectively, the numbers in parentheses show the p values. Al abbreviations and sources are the same as in Table 1 with the authors' calculations. In FM-OLS, lags 2.00 are used, and the linear and quadratic eqtrend effect is controlled (2).

Table 9
Direct and indirect effects of corruption (CCI) on economic growth. FM-OLS estimation.

	CCI DIRECT EFFECT	CHANNEL TO LGDP	CCI TO CHANNEL	INDIRECT EFFECT	TOTAL EFFECT	INDIRECT/ TOTAL	INDIVIDUAL MODELS
INVT	-0,030850	0,023657	-0,235138	-0,005563	-0,036413	15,28%	**CHANNEL 1**
SEDU	-0,016625	0,021421	-0,698978	-0,014973	-0,031598	47,39%	**CHANNEL 2**
RDEXP	-0,020245	0,256387	-0,034018	-0,008722	-0,028967	30,11%	**CHANNEL 3**
		FINAL MODEL					
CCI		-0,010485				31,16%	**FINAL MODEL**
INVT		0,013518	-0,235138	-0,003179	0,010339	9,44%	**CHANNEL 1**
SEDU		0,019028	-0,698978	-0,013300	0,005728	39,52%	**CHANNEL 2**
RDEXP		0,196636	-0,034018	-0,006689	0,189946	19,88%	**CHANNEL 3**
			TOTAL	-0,033653		100,00%	
			WITHOUT CHANNELS	-0,032360		**RELATIVE SIGNIFICANCE**	

All abbreviations and sources are the same as in Table 1 with the authors' calculations.

Table 9 shows the total results of all previous econometric calculations. The first part of the table records the results on each channel separately and the estimates through the Eqs. (2). We observe that the effect of corruption on growth through the individual investment channel is 15.28% of the total channels of the model, which is quite significant. Moreover, the impact from the individual channels of human capital and innovation is 47.39% and 30.11%, respectively. At the same time, the channels of the model that are not included constitute the remaining 7.22%, implying that the analysis is robust. Meanwhile, the second part of the table showcases the overall model.

This overall result is mainly attributed to the indirect effects of corruption on human capital, total investment, and innovation, corresponding to 39.52%, 9.44%, and 19.88%, respectively, of the overall effect of corruption on growth. The three channels collectively account for 68.84% of the total impact of corruption, while the direct effect of corruption on growth is 31.16%.

In general, corruption reduces productivity. If corruption increases at a greater rate than productivity, resources will flow from productive activities to corrupt activities over time (Mo, 2001). In corrupt countries, this results in a lower stock of productive inputs, such as investment, human capital, and innovative activities. The above analysis helps explain the mechanisms through which corruption affects economic growth. While human capital is the most vital channel of influence, innovation cannot be ignored. Because the literature has not sufficiently analysed the effects of innovation, we conduct an in-depth examination of the negative correlation between corruption and economic growth in the next section. We note that this impact is significant and differentiates policy actions toward tackling corruption, which is the focus of this article[2].

4.4. Empirical results of the impact of innovation on economic growth through corruption

The effect of innovation on the corruption-growth nexus is significant and must be studied in depth (lines f and g of Fig. 1). Table 10 reveals the correlations through the various econometric models. Eq. (1) introduces the variable

[2] Appendix D presents the robustness check for transmission variables (Table D.1).

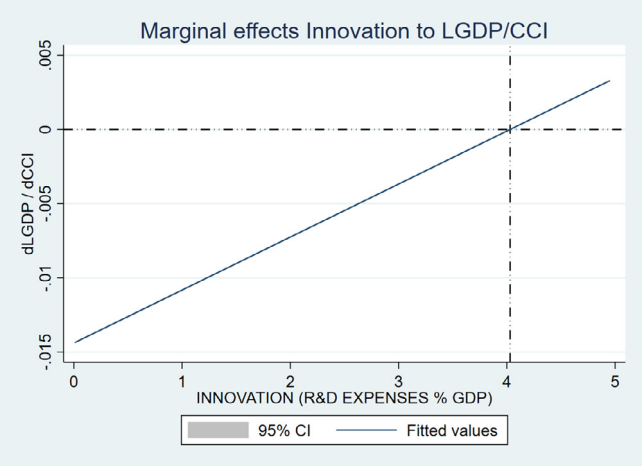

Fig. 2. Diagram shows the marginal effects of innovation on the impact of corruption in economic growth. Source: Equation (4) with data from sources: World Bank, IMF, Global Economy, OECD, UNCTAD, Transparency International, and authors' calculations.

CCIRD, showing the combined effect of innovation through corruption (CCI * RD) on economic growth. Moreover, the variables of corruption (CPI and ICRG) and innovation (TRMRK and PAT) have been introduced to investigate the validity of econometric data. Additionally, for the econometric procedures, the variables CCIPAT, CCPITRMRK, CPIRD, CPIPAT, CPITRMRK, ICRGRDEXP, ICRGPAT, and ICRGTRMRK have been introduced, which show the combined activities of corruption with innovation.

The model shows the impact of innovation on economic growth directly and through corruption. It shows coherence and that an increase in RDEXP of 0.1% of GDP in Model 1 increases GDPpc by 1.97%, thereby confirming the literature findings in an extensive sample of countries. In addition, the combined factor of the simultaneous impact of innovation on economic growth through corruption (CCIRD) offers valuable insights.

Deriving Eq. (1), in terms of corruption, results in the following:

$$\frac{\partial LGDP}{\partial CCI} = (\text{coefficient CCI}) + (\text{coefficient CCIRD}) * \text{RDEXP} \tag{3}$$

Replacing the coefficient of the FM-OLS method, we end up with the following equation:

$$\frac{\partial LGDP}{\partial CCI} = (-0.014394) + (0.003573) * \text{RDEXP} \tag{4}$$

Eq. (4) shows that corruption does not affect all countries equally when innovation increases. In particular, improving innovation reduces the negative impact of corruption on economic growth. Suppose we were to solve the above equation by determining its reversal point. In that case, we observe that at the value RDEXP = 4.03 (% of GDP) (Israel and South Korea), we would find a zeroing of the effect of corruption on economic growth. The results are shown in Fig. 2. The hypothesis under investigation is whether the expansion of innovative activities can weaken corruption in terms of its impact on economic growth. That will only take place if the competitive field created by enhancing innovation forces companies to turn to highly productive activities to survive. Such a competitive process would make corruption a deterrent to overall business activity, increasing uncertainty and risk instead of reducing it. Countries can reduce the harmful effects of corruption on economic growth concerning policy decisions by enhancing innovation. It is also notable that boosting innovation is generally easier than fighting corruption, which is a complex phenomenon, and its relationship with the economy and society per se remains highly problematic.

An analysis of the correlation between innovation and corruption reveals that changing corruption by one unit reduces innovation by about 4%, which was expected based on the theory and the literature[3].

5. Conclusion

This study estimated the long-term relationship between corruption and economic growth in 109 countries from 2010 to 2018. The empirical model is the starting point of our research by introducing growth model variables. The strong negative relationship between corruption and economic growth according to the "sand the wheels" hypothesis was confirmed. An increase in corruption by one unit reduces GDPpc by 1.3%. The indirect impact of corruption through

[3] In Appendix E, we present the robustness check for the innovation-corruption-economic growth model (Table D.1, Fig. A).

Table 10
Direct and indirect effects of corruption (CCI) on economic growth. FM-OLS estimation.

Dependent variable	Logarithm of GDP per capita: LGDP 981 observations in 109 countries (total)								
Model	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	MODEL 9
CCI	−0.014394*** (0.000)	−0.021058*** (0.000)	−0.019908*** (0.000)						
CPI				−0.014646*** (0.000)	−0.019089*** (0.000)	−0.017727*** (0.000)			
ICRG							−0.248405*** (0.000)	−0.299710*** (0.000)	−0.253907*** (0.001)
POP	−0.141788*** (0.000)	−0.098560*** (0.001)	−0.152348** (0.011)	−0.132105*** (0.000)	−0.106626*** (0.000)	−0.117983** (0.017)	−0.127216*** (0.000)	−0.105626*** (0.000)	−0.109506* (0.069)
SEDU	0.017377*** (0.000)	0.017232*** (0.000)	0.016450*** (0.000)	0.018761*** (0.000)	0.017932*** (0.000)	0.019110*** (0.000)	0.019084*** (0.000)	0.020940*** (0.000)	0.020234*** (0.000)
INVT	0.013042*** (0.000)	0.007277* (0.076)	0.014006 (0.110)	0.013844*** (0.000)	0.010325** (0.011)	0.009672 (0.182)	0.011451*** (0.000)	0.009878** (0.016)	0.016047* (0.072)
TROP	0.002702*** (0.000)	0.000482 (0.351)	0.002319** (0.034)	0.002352*** (0.000)	0.000517 (0.309)	0.001869** (0.038)	0.002915*** (0.000)	0.001233** (0.017)	0.002166** (0.046)
BRM	0.000721*** (0.002)	0.001165** (0.032)	0.001088 (0.293)	0.000946*** (0.000)	0.000992* (0.061)	0.001668* (0.051)	0.000589*** (0.006)	0.000207 (0.694)	0.001519 (0.149)
GEXP	−0.033767*** (0.000)	−0.016156*** (0.000)	−0.021078** (0.032)	−0.032852*** (0.000)	−0.019784*** (0.000)	−0.020973*** (0.010)	−0.039726*** (0.000)	−0.025172*** (0.000)	−0.020861** (0.036)
RDEXP	0.120918*** (0.000)			0.030914 (0.401)			0.122208*** (0.000)		
CCIRD	0.003573*** (0.000)								
TRMRK		0.0000091 (0.750)			0.0000119*** (0.730)			0.0000171 (0.585)	
CCITRMRK		0.00000601*** (0.000)							
PAT			0.000300 (0.227)			0.0003324 (0.163)			0.0002151 (0.486)
CCIPAT			0.0000138* (0.071)						
CPIRD				0.004810*** (0.000)					
CPITRMRK					0.00000487*** (0.000)				
CPIPAT						0.0000128** (0.042)			
ICRGRD							−0.049668*** (0.000)		
ICRGTRMRK								0.0000825*** (0.000)	
ICRGPAT									0.0001586 (0.200)
R^2	0.3507	0.3367	0.4208	0.3562	0.3137	0.4336	0.3722	0.2825	0.4641

CCIRD, CCITRMRK, CCIPAT, CPIRD, CPITRMRK, CPIPAT, ICRGRD, ICRGTRMRK and ICRGPAT are cross term variables. *, ** and *** mean the significance level of 10%, 5% and 1% respectively, the numbers in parentheses show the p values. All other abbreviations and sources are the same as in Table 1 with authors' calculations. In FM-OLS, lags 2.00 are used, and the linear and quadratic eqtrend effect is controlled (2).

the human capital channel was 39.52%, investment 9.44%, innovation 19.88%, and direct effect 31.16%. The differences between developing and developed countries and those across geographic regions were also researched, with interesting policy implications. By studying the impact of innovation on the relationship between corruption and economic growth, we identified the countries in which the negative effect of corruption is mitigated. The findings imply that it is easier to strengthen innovation at the level of political action than to directly fight corruption (a more complex and challenging process, as it is also linked to an economy's social structure and culture). This was done for two reasons: (1) reaping the immediate benefits of innovative activity by confirming the conclusions of UNESCO's Sustainable Development Goals Report of 2018, which associated reduced amounts spent only by 0.1 to 0.6% of their GDP on R&D activities with low economic growth; and (2) reducing the negative effect of corruption, diminishing its demand, and resulting in its lower supply.

The findings were achieved through the PCSE models and the study of autocorrelation, heteroscedasticity, and endogeneity before ending with the FM-OLS methodology. In addition, various variables on corruption (CCI, CPI, and ICRG) and innovation (R&D expenses, trademarks, and patents), many of which have different structures and origins, confirmed the results. Moreover, robustness checks divided our model into developmental categories and regions, minimising estimation bias and geographical consistency and, identifying the potential differences between regions.

Next, we clarified corruption, innovation, and economic growth interactions. Because corruption impacts innovation and economic growth, we start there. Our research controlled the innovation channel through variables. The study's goal was to determine how much of corruption's impact on innovation is transferred to GDP per capita. Finally, we examined whether and how innovation mitigates corruption's negative effects on economic growth. This study used cross-term factors to show how innovation affects the corruption-growth nexus. We studied how this effect changes, its thresholds, and the countries with the strongest effects.

In future research, we suggest extending the research periods. As a result, the Granger causality approach could be utilised more robustly to study the causal linkages between the model variables. In addition, more transmission channels could also be tested regarding the effects of corruption and the causes that affect economic growth, such as the FDI, political stability, and type of government.

Acknowledgement

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Appendix A. Dumitrescu–Hurlin causality test for panel data.

Dumitrescu and Hurlin (2012) proposed an extension of the Engle and Granger (1987) model, who argued that the proof of the occurrence of an integrated vector between variables might be followed by a causal relationship between them in one or both directions to identify causal relationships in panel data. The introduced correlation is:

$$y_{\mathrm{i,t}} = a_i + \sum_{k=1}^{K} \gamma_{i,k} y_{i,t-k} + \sum_{k=1}^{K} \beta_{ik} \chi_{i,t-k} + \varepsilon_{\mathrm{i,t}}, t = 1, 2, \ldots, \mathrm{T} \tag{A}$$

In this equation, $\gamma_{i,k}$ and $x_{i,k}$ are stationary series of i country in time t. Estimates vary from country to country but are time-varying. Time K is the same for all units, and the panel is considered balanced. $\varepsilon_{\mathrm{i,t}}$ is the usual disturbance terms with zero means and finite variances. For simplicity, individual effects a_i are assumed to be fixed. In addition, lag orders are considered identical for all cross-section units of the panel. Past values of x are considered as estimators of y. Again, the causality is checked by Eq. (A). No test was performed to determine the time lags because we can apply the test only for one period (lag (K) = 1), as we transformed the data for nine periods (T> 5 + 3K).

Appendix B. Descriptive statistics

See Table B.1.

Appendix C. Unit root tests.

See Table C.1.

The results highlight that GDP per capita, corruption (all three indicators), total investment, human capital, labour, trade openness, broad money, innovation (all indicators), and government size are stationary in the first differences. The null hypothesis of non-stationarity is rejected.

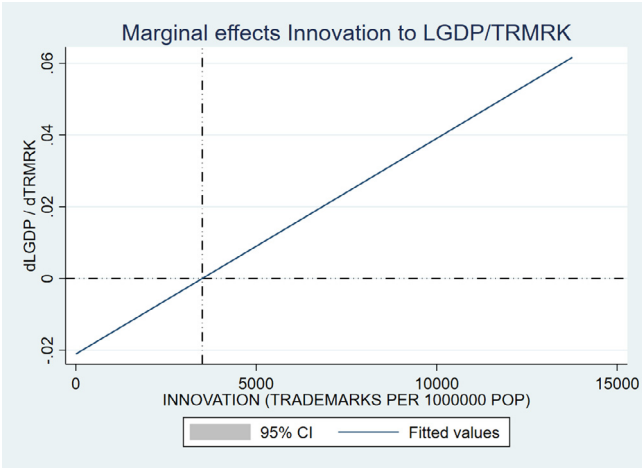

Fig. A. Diagram shows the marginal effects of innovation on the impact of corruption in economic growth. Source: Eq. (C) with data from sources: World Bank, IMF, Global Economy, OECD, UNCTAD, Transparency International, and authors' calculations.

Table B.1
Statistical characteristics of the variables of the aggregate sample.

Variables	Observations	Min $	Max $	Mean Value $	Standard deviation
GDPpc	981 (109 countries)	659,80	93.940,80	20.784,24	18.411,67
LGDP	981 (109 countries)	6,49	11,45	9,45	1,13
CPI	981 (109 countries)	5	89	52,73	20,51
CCI	981 (109 countries)	1,90	83,46	46,91	20,84
ICRG	981 (109 countries)	0	5,62	3,17	1,22
POP	981 (109 countries)	−2,26	7,69	1,20	1,20
SEDU	981 (109 countries)	23,29	163,93	87,50	28,32
INVT	981 (109 countries)	8,95	48,41	22,84	6,35
TROPEN	981 (109 countries)	0,20	442,62	91,56	64,30
BRM-M3/GDP	981 (109 countries)	10,11	699,20	76,10	73,82
GEXP	981 (109 countries)	4,40	73,58	16,32	6,04
RDEXP	981 (109 countries)	0,0085	4,95	0,92	0,98
TRMRK	981 (109 countries)	1,04	13759,36	1.306,50	1.788,14
PAT	981 (109 countries)	0,039	4.188,85	236,60	561,46

Table B.1: Descriptive statistics of all model variables for the whole sample of countries according to the databases mentioned and personal calculations.

Appendix D. Robustness check in transmission variables

Correspondingly, we can summarise the effect in a model with a different corruption index. The ICRG index was chosen because it has a significantly different structure and composition from the CCI than the CPI, which has the same design. Below is the summary table of the corresponding effects (Table D.1), which shows many similarities with the CCI, thus confirming our findings. The most crucial difference is that the investment channel (17.59%) has a more significant impact on the effect of corruption on growth than the CCI model (9.44%), detaching an impact from the direct effect of corruption on growth (23.44% from 31.16%). The other effects are similar.

Appendix E. Robustness check in innovation-corruption-growth model

We introduce the two differently structured innovation variables to test the model's resilience to change: trademarks (TRMRK) and patents (PAT). Additionally, we introduce variables of corruption, namely CPI and ICRG. The correlation analysis is shown with corresponding effects in Table 10 (Model 2). We observe that an increase in trademarks by 1 per 100 inhabitants causes an increase in GDPpc by 0.9%. In addition, the combined factor of the simultaneous impact of innovation on economic growth through corruption (CCITRMRK) offers valuable conclusions (see Table D.1).

Deriving Eq. (1), in terms of corruption, results in the following:

$$\frac{\partial LGDP}{\partial CCI} = (\text{coefficient CCI}) + (\text{coefficient CCITRMRK}) * \text{TRMRK} \tag{B}$$

Table C.1
LLC, BREITUNG, and ADF-FISHER unit roots test at levels and first differences.

Variables	LLC		BREITUNG		ADF-FISHER	
	Levels	1st difference	Levels	1st difference	Levels	1st difference
LGDP	−6,805***	−23,614***	14,911	−3,060***	2,058	−10,238***
	(0,000)	(0,000)	(1,000)	(0,001)	(0,980)	(0,000)
CPI	−46,052***	−24,662***	−0,194	−12,507***	−3,157***	−20,175***
	(0,000)	(0,000)	(0,432)	(0,000)	(0,0008)	(0,000)
CCI	−15,768***	−22,243***	0,150	−10,910***	0,938	−13,110***
	(0,000)	(0,000)	(0,560)	(0,000)	(0,826)	(0,000)
ICRG	−100***	−35,391***	1,431	−8,840***	−1,783**	−10,503***
	(0,000)	(0,000)	(0,924)	(0,000)	(0,037)	(0,000)
POP	−15,516***	−46,679***	6,627	2,468	−2,506***	−8,519***
	(0,000)	(0,000)	(1,000)	(0,993)	(0,006)	(0,000)
SEDU	−21,171***	−57,639***	4,499	−7,881***	−2,882***	−13,615***
	(0,000)	(0,000)	(1,000)	(0,000)	(0,002)	(0,000)
INVT	−18,171***	−27,000***	0,204	−7,841***	−3,462***	−13,710***
	(0,000)	(0,000)	(0,581)	(0,000)	(0,000)	(0,000)
TROP	−10,895***	−14,157***	1,991	−6,844***	−0,405	−16,099***
	(0,000)	(0,000)	(0,977)	(0,000)	(0,347)	(0,000)
BRM	−6,813***	−18,600***	5,300	−8,093***	4,520	−15,189***
	(0,000)	(0,000)	(1,000)	(0,000)	(1,000)	(0,000)
GEXP	−13,493***	−23,911***	2,008	−7,225***	−0,104	−16,349***
	(0,000)	(0,000)	(0,978)	(0,000)	(0,459)	(0,000)
GII	−27,626***	−29,119***	−3,537***	−17,899***	−1,438*	−18,598***
	(0,000)	(0,000)	(0,000)	(0,000)	(0,075)	(0,000)
RDEXP	−0,5120	−34,120***	6,008	−9,528***	4,753	−15,487***
	(0,2901)	(0,000)	(1,000)	(0,000)	(1,000)	(0,000)
TRMRK	−12,102***	−7,726***	5,910	−7,574***	−4,685***	−21,695***
	(0,000)	(0,000)	(1,000)	(0,000)	(0,000)	(0,000)
PAT	−6,243***	−22,989***	1,887	−9,463***	−1,308*	−19,025***
	(0,000)	(0,000)	(0,970)	(0,000)	(0,095)	(0,000)

The table presents the panel and 1st difference unit root results for 109 countries. In all tests, the null hypothesis is that there is a non-stationarity. The numbers in parentheses indicate the p-values. *, **, *** symbolises the rejection of the null hypothesis at a significance level of 10%, 5%, and 1%, respectively. Al abbreviations and sources are the same as in Table 1.

Table D.1
Direct and indirect effects of corruption (ICRG) on economic growth; FM-OLS estimation method.

	ICRG DIRECT EFFECT	CHANNEL TO LGDP	ICRG TO CHANNEL	INDIRECT EFFECT	TOTAL EFFECT	INDIRECT /TOTAL	INDIVIDUAL MODELS
INVT	-0,495877	0,022992	-9,064168	-0,208406	-0,704283	29,59%	**CHANNEL 1**
SEDU	-0,253941	0,024803	-11,218620	-0,278252	-0,532193	52,28%	**CHANNEL 2**
RDEXP	-0,387226	0,239182	-0,526726	-0,125983	-0,513209	24,55%	**CHANNEL 3**
		FINAL MODEL					
ICRG		-0,142987				23,44%	**FINAL MODEL**
INVT		0,011841	-9,064168	-0,107324	-0,095484	17,59%	**CHANNEL 1**
SEDU		0,022253	-11,218620	-0,249649	-0,227396	40,92%	**CHANNEL 2**
RDEXP		0,208972	-0,526726	-0,110071	0,098901	18,04%	**CHANNEL 3**
			TOTAL	-0,610031		100,00%	
			WITHOUT CHANNELS	-0,502642		**RELATIVE SIGNIFICANCE**	

Al abbreviations and sources are the same as in Table 1 with the authors' calculations.

Replacing the coefficient of the FM-OLS method, we end up with the equation:

$$\frac{\partial LGDP}{\partial CPI} = (-0.021058) + (0.00000601) * \text{TRMRK} \tag{C}$$

Eq. (C) shows that improving innovation reduces the negative impact of corruption on economic growth. A zeroing effect of corruption on economic growth is found at the value TRMRK = 3504. This result is in line with Model 1, as illustrated

Table F.1
Direct and indirect effects of corruption (CCI) on economic growth. FM-OLS estimation.

	CCI direct effect	Channel to LGDP	CCI to channel	Indirect effect	Total effect	Indirect/Total	Individual models
INVT	−0,030850	0,023657	−0,235138	−0,005563	−0,036413	15,28%	CHANNEL 1
SEDU	−0,016625	0,021421	−0,698978	−0,014973	−0,031598	47,39%	CHANNEL 2
RDEXP	−0,020245	0,256387	−0,034018	−0,008722	−0,028967	30,11%	CHANNEL 3
		FINAL MODEL					
CCI		−0,010485				31,16%	FINAL MODEL
INVT		0,013518	−0,235138	−0,003179	0,010339	9,44%	CHANNEL 1
SEDU		0,019028	−0,698978	−0,013300	0,005728	39,52%	CHANNEL 2
RDEXP		0,196636	−0,034018	−0,006689	0,189946	19,88%	CHANNEL 3
			TOTAL	−0,033653		100,00%	
			WITHOUT CHANNELS	−0,032360		RELATIVE SIGNIFICANCE	

All abbreviations and sources are the same as in Table 1 with the authors' calculations.

Table F.2
Direct and indirect effects of corruption (ICRG) on economic growth. FM-OLS estimation method.

	ICRG direct effect	Channel to LGDP	ICRG to channel	Indirect effect	Total effect	Indirect/Total	Individual models
INVT	−0,495877	0,022992	−9,064168	−0,208406	−0,704283	29,59%	CHANNEL 1
SEDU	−0,253941	0,024803	−11,218620	−0,278252	−0,532193	52,28%	CHANNEL 2
RDEXP	−0,387226	0,239182	−0,526726	−0,125983	−0,513209	24,55%	CHANNEL 3
		FINAL MODEL					
ICRG		−0,142987				23,44%	FINAL MODEL
INVT		0,011841	−9,064168	−0,107324	−0,095484	17,59%	CHANNEL 1
SEDU		0,022253	−11,218620	−0,249649	−0,227396	40,92%	CHANNEL 2
RDEXP		0,208972	−0,526726	−0,110071	0,098901	18,04%	CHANNEL 3
			TOTAL	−0,610031		100,00%	
			WITHOUT CHANNELS	−0,502642		RELATIVE SIGNIFICANCE	

Al abbreviations and sources are the same as in Table 1 with the authors' calculations.

in Fig. A. The only difference is that it increases the sample of countries where innovation eliminates the adverse effects of corruption (Sweden, the Netherlands, Switzerland, France, Japan, South Korea, Hong Kong, New Zealand, Australia, and Bahrain).

Most models are shown to behave similarly, as the relationship is similar and shows a reverse point in RDEXP = 3.04% GDP in CPI (Model 4), RDEXP = 5.01% GDP in ICRG (Model 7), PAT = 1442 per 1,000,000 inhabitants in CCI (Model 3), PAT = 1384 per 1,000,000 inhabitants in CPI (Model 6). TRMRK = 3503 per 1,000,000 inhabitants in CCI (Model 2), TRMRK = 3915 per 1,000,000 inhabitants in CPI (Model 5), and TRMRK = 3633 per 1,000,000 inhabitants in ICRG (Model 8).

Appendix F. The 109 countries used in this study

Albania, United Arab Emirates, Argentina, Armenia, Australia, Austria, Azerbaijan, Burundi, Belgium, Bulgaria, Bahrain, Bosnia and Herzegovina, Belarus, Bolivia, Brazil, Botswana, Canada, Switzerland, Chile, China, Cote d'Ivoire, Congo Dem. Rep., Colombia, Cabo Verde, Costa Rica, Czech Republic, Germany, Denmark, Algeria, Ecuador, Egypt, Spain, Estonia, Ethiopia, Finland, France, Gabon, United Kingdom, Ghana, Gambia, Greece, Guatemala, Hong Kong, Honduras, Croatia, Hungary, Indonesia, India, Ireland, Iceland, Israel, Italy, Jamaica, Jordan, Japan, Kazakhstan, Kenya, Korea Rep, Lao PDR, Sri Lanka, Lithuania, Luxembourg, Latvia, Morocco, Moldova, Madagascar, Mexico, Mali, Malta, Myanmar, Mongolia, Mozambique, Malawi, Malaysia, Namibia, Nigeria, Nicaragua, Netherlands, Norway, New Zealand, Pakistan, Panama, Peru, Philippines, Poland, Portugal, Paraguay, Romania, Russia, Sudan, Senegal, Singapore, El Salvador, Serbia, Slovak Republic, Slovenia, Sweden, Togo, Thailand, Tunisia, Turkey, Tanzania, Uganda, Ukraine, Uruguay, United States, South Africa, Zambia, Zimbabwe.

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